Galilean and Lorentz Transformations

PHYS 2203 — Introductory Physics III

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Overview

This lecture introduces the concept of **coordinate transformations**. We begin with the **Galilean transformations** of classical physics and see precisely why they cannot keep the speed of light fixed. Imposing the two postulates of special relativity yields the **Lorentz transformations**, the single rule relating the spacetime coordinates of an event in two frames in uniform relative motion. From them we recover *length contraction* and *time dilation* in terms of the *Lorentz factor* γ, establish the *relativity of simultaneity*, and check that everything reduces to Newtonian physics in the classical limit $v \ll c$.

1 Classical (Galilean) Relativity

Consider two inertial frames in the standard configuration: an unprimed frame O and a primed frame O' that moves at constant velocity v along the common x-axis. Throughout, we reserve v for the relative velocity *between the two frames*, and use u for the velocity of an *object* measured within a frame. After a time t, the origin of O' has moved a distance vt relative to O.

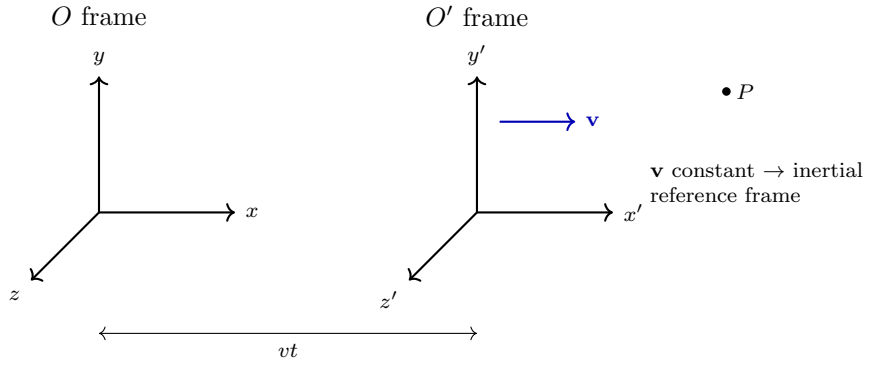


Figure 1: Standard configuration. An event at point P described by two inertial observers. Frame O' moves with constant velocity $\mathbf{v}$ along x, so after a time t its origin has shifted by vt relative to O.

An event at P has coordinates (x, y, z, t) in O and (x', y', z', t') in O'. Classically these are related

by the **Galilean transformations**:

$$\boxed{\begin{aligned} x' &= x - vt, \\ y' &= y, \\ z' &= z, \\ t' &= t. \end{aligned}} \tag{1}$$

Notice the last line: classically, time is assumed *absolute*, the same in every frame.

Classical velocity addition

Differentiating $x' = x - vt$ in time relates the velocity of an object in the two frames. If the object moves at velocity u along x as measured in O, then in O' it moves at

$$\boxed{u' = u - v} \tag{2}$$

(more generally, $u'_x = u_x - v$, $u'_y = u_y$, $u'_z = u_z$). This is the **classical velocity-addition** rule. It should apply to particles, to objects, and — classically — to waves such as light.

The conflict with the speed of light

If light moves at $u = c$ in frame O, the classical rule predicts a different speed $u' = c - v$ in O'. But Einstein's second postulate — confirmed by the null result of Michelson–Morley — demands that light travel at exactly c in *every* inertial frame. The classical velocity-addition rule, and with it the Galilean transformation, cannot be correct. We need a transformation that keeps the speed of light fixed in all frames: the **Lorentz transformation**.

2 Deriving the Lorentz Transformations

We keep the standard configuration of Figure 1: frame O at rest, frame O' moving at velocity v along the common x/x' axis, and the origins coinciding at $t = t' = 0$.

Step 1: the transformation must be linear

A free object — one with no force on it — moves in a straight line at constant velocity in *any* inertial frame. That is what makes a frame inertial in the first place. So the transformation between two inertial frames must carry straight-line, constant-velocity motion into straight-line, constant-velocity motion.

Only a **linear** relation does that. A term in x^2, t^2, or xt would bend uniform motion into accelerated motion and destroy the inertial character of the second frame. The most general linear relation between the two sets of coordinates is therefore

$$\boxed{\begin{aligned} x' &= A\,x + B\,t, \\ t' &= C\,x + D\,t, \end{aligned}} \tag{3}$$

where A, B, C, D are constants — they may depend on the relative speed v, but not on x or t. (Anything else would not be linear.)

We now have four unknowns, and we will determine them with four physical conditions. Note that the Galilean transformation is the special case $A = 1$, $B = -v$, $C = 0$, $D = 1$; the interesting physics is in how the true values differ.

Step 2: the origin of O' fixes B

In frame O the origin of O' sits at $x = vt$. In frame O' that same origin is at $x' = 0$ by definition. Substituting both into the first equation,

$$0 = A\,(vt) + B\,t = (Av + B)\,t. \tag{4}$$

This must hold at every time t, so the bracket vanishes:

$$\boxed{B = -A\,v} \qquad \Longrightarrow \qquad x' = A\,(x - vt). \tag{5}$$

Step 3: the origin of O fixes D

Now run the same argument the other way. The origin of O sits at $x = 0$. Putting $x = 0$ into both equations gives its description in O':

$$x' = B\,t = -Av\,t, \qquad t' = D\,t. \tag{6}$$

Its velocity as measured in O' is therefore

$$\frac{x'}{t'} = \frac{-Avt}{Dt} = -\frac{A}{D}\,v. \tag{7}$$

But seen from O', frame O recedes in the $-x'$ direction at speed exactly v — this is the principle of relativity, which forbids the two frames from disagreeing about their mutual speed. Hence $-\frac{A}{D}v = -v$, giving

$$\boxed{D = A} \qquad \Longrightarrow \qquad t' = C\,x + A\,t. \tag{8}$$

Step 4: the second postulate fixes C

So far we have used only linearity and the relative motion of the origins. Now bring in the constancy of the speed of light.

Let a light pulse leave the common origin at $t = t' = 0$ in the $+x$ direction. The second postulate says *both* observers measure its speed to be c:

$$x = c\,t \quad \text{in } O, \qquad\qquad x' = c\,t' \quad \text{in } O'. \tag{9}$$

Substituting $x = ct$ into our two equations,

$$x' = A\,(ct - vt) = A\,t\,(c - v), \tag{10}$$

$$t' = C\,(ct) + A\,t = t\,(Cc + A). \tag{11}$$

Now impose $x' = c\,t'$:

$$\begin{aligned} A\,t\,(c - v) &= c\,t\,(Cc + A) \\ A\,(c - v) &= Cc^2 + Ac \\ Ac - Av &= Cc^2 + Ac \\ -Av &= Cc^2. \end{aligned} \tag{12}$$

Therefore

$$\boxed{C = -\frac{Av}{c^2}} \qquad \Longrightarrow \qquad t' = A\Big(t - \frac{v}{c^2}\,x\Big)\,. \tag{13}$$

Three constants are now expressed in terms of the fourth. Collecting what we have,

$$x' = A\,(x - vt), \qquad t' = A\Big(t - \frac{v}{c^2}\,x\Big)\,. \tag{14}$$

Step 5: the principle of relativity fixes A

One constant remains, and one physical principle is left unused. Invert the pair of equations above — solve them for x in terms of the primed coordinates. Multiply the second by v and add it to the first:

$$\begin{aligned} x' + v\,t' &= A\,(x - vt) + Av\Big(t - \frac{v}{c^2}x\Big) \\ &= A\left[\,x - vt + vt - \frac{v^2}{c^2}x\,\right] \\ &= A\,x\left(1 - \frac{v^2}{c^2}\right), \end{aligned} \tag{15}$$

so that

$$x = \frac{1}{A\left(1 - \dfrac{v^2}{c^2}\right)}\,\left(x' + v\,t'\right). \tag{16}$$

Here is the decisive step. Neither frame is privileged, so the transformation from O' back to O must look exactly like the transformation from O to O', with the sign of v reversed. That means the constant multiplying $(x' + vt')$ must be the *same* A:

$$\frac{1}{A\left(1 - \dfrac{v^2}{c^2}\right)} = A. \tag{17}$$

Solving,

$$\begin{aligned} A^2\left(1 - \frac{v^2}{c^2}\right) &= 1 \\ A^2 &= \frac{1}{1 - v^2/c^2}, \end{aligned} \tag{18}$$

and therefore

$$\boxed{A = \frac{1}{\sqrt{1 - v^2/c^2}} \equiv \gamma} \tag{19}$$

— the Lorentz factor of Lecture 11. There it was *defined* from the light clock; here it emerges as the only constant consistent with the postulates. That the same number arrives by two independent routes is a real check on the theory.

The perpendicular coordinates

Nothing above touched y or z, and in fact $y' = y$ and $z' = z$: lengths perpendicular to the motion are unchanged.

The reason is a symmetry argument. Imagine two identical rings, one at rest in each frame, sliding toward each other along their common axis. If transverse lengths shrank with motion, then O would say the moving ring passes *inside* his, while O' would say exactly the reverse. Whether one ring passes inside the other is a physical fact both must agree on, so the only consistent possibility is that transverse lengths are unaffected.

The Lorentz transformation equations

Collecting the four results, the Lorentz transformation and its inverse are

$$\boxed{\begin{aligned} x' &= \gamma\,(x - vt) \\ y' &= y \\ z' &= z \\ t' &= \gamma\big[\,t - \tfrac{v}{c^2}\,x\,\big] \end{aligned}} \qquad \text{or} \qquad \boxed{\begin{aligned} x &= \gamma\,(x' + vt') \\ y &= y' \\ z &= z' \\ t &= \gamma\big[\,t' + \tfrac{v}{c^2}\,x'\,\big] \end{aligned}} \tag{20}$$

The inverse is obtained, as it must be, by exchanging primed and unprimed coordinates and reversing the sign of v. Note also that the transformation keeps light at speed c by *construction* — that was the input at Step 4, not a coincidence to be checked afterwards.

For *intervals* between two events, the same linear relations hold for the differences (the transformation is linear, so constants subtract away):

$$\Delta x' = \gamma\,(\Delta x - v\,\Delta t), \qquad \Delta t' = \gamma\Big(\Delta t - \frac{v}{c^2}\,\Delta x\Big)\,. \tag{21}$$

3 Consequences of the Lorentz Transformations

We now collect the interval forms of the transformations and read off their physical consequences. For an event 1 at (x_1, y_1, z_1, t_1) and an event 2 at (x_2, y_2, z_2, t_2), define $\Delta x = x_2 - x_1$ and $\Delta t = t_2 - t_1$. Then

$$\Delta x' = \gamma\,(\Delta x - v\,\Delta t), \quad \Delta y' = \Delta y, \quad \Delta z' = \Delta z, \quad \Delta t' = \gamma\Big(\Delta t - \frac{v}{c^2}\,\Delta x\Big)\,. \tag{22}$$

Length contraction

To measure the length of a moving object, one records the positions of its two edges. Let event 1 be the measurement of the left edge and event 2 the measurement of the right edge.

Suppose the object is at rest in O', with **rest length** (also called proper length)

$$L_0 = x'_2 - x'_1. \tag{23}$$

In the O frame the object is moving, so its two edges must be measured *simultaneously*: $t_1 = t_2$, hence $\Delta t = 0$. Applying the transformation,

$$\Delta x' = \gamma\,(\Delta x - v\,\Delta t) \quad \Longrightarrow \quad L_0 = \gamma\,(L + 0) \quad \Longrightarrow \quad \boxed{L = \frac{L_0}{\gamma} < L_0.} \tag{24}$$

Here L is the length measured when the object is moving, and L_0 is the rest length. A moving object is measured to be shorter than its rest length.

Time dilation

Let event 1 be "clock ticks" and event 2 be "clock ticks again." Suppose the clock is at rest in O', so $x'_1 = x'_2$ and $\Delta x' = 0$. The O' observer, at rest relative to the clock, measures the **proper time** $\Delta t' = \Delta t_0$.

In the O frame the clock has moved a distance $v\,\Delta t$ between ticks. Using the inverse transformation,

$$\Delta t' = \gamma\Big(\Delta t - \frac{v}{c^2}\,\Delta x\Big) \implies \Delta t_0 = \gamma\Big(\Delta t - \frac{v}{c^2}\,v\,\Delta t\Big)$$
$$= \gamma\,\Delta t\left(1 - \frac{v^2}{c^2}\right) = \frac{\Delta t}{\gamma}, \tag{25}$$

using $1 - v^2/c^2 = 1/\gamma^2$. Solving for the interval measured in the frame where the clock is moving,

$$\boxed{\Delta t = \gamma\,\Delta t_0 > \Delta t_0.} \tag{26}$$

The time interval measured when the clock is moving (Δt) exceeds the proper time (Δt_0): moving clocks run slow.

Relativity of simultaneity

Suppose two events occur *simultaneously* according to O, so $\Delta t = 0$. What does O' measure? Use the *inverse* transformation, whose left-hand side is the unprimed interval,

$$\Delta t = \gamma\Big(\Delta t' + \frac{v}{c^2}\,\Delta x'\Big)\,, \tag{27}$$

and set $\Delta t = 0$:

$$0 = \gamma\Big(\Delta t' + \frac{v}{c^2}\,\Delta x'\Big) \implies \boxed{\Delta t' = -\frac{v}{c^2}\,\Delta x'.} \tag{28}$$

Note that γ cancels entirely. The offset vanishes only when $\Delta x' = 0$, that is, only when the two events happen at the same place *along the direction of motion*. Therefore:

$$\boxed{\text{Two events simultaneous in } O \text{ are simultaneous in } O' \text{ only if } \Delta x = 0.}$$

Events separated only in y or z stay simultaneous — it is separation along the boost direction that destroys simultaneity.

The classical limit $v \ll c$

Every relativistic effect above is controlled by the Lorentz factor γ. At everyday speeds $v \ll c$, so $v^2/c^2 \approx 0$ and

$$\gamma \longrightarrow 1. \tag{29}$$

In this limit the consequences switch off one by one:

- **Length contraction:** $L = L_0/\gamma \to L_0$. Moving objects keep their rest length.
- **Time dilation:** $\Delta t = \gamma\,\Delta t_0 \to \Delta t_0$. Clocks run at the same rate in every frame.
- **Simultaneity:** the offset $(v/c^2)\,\Delta x \to 0$. Simultaneity becomes absolute.

The Lorentz transformation itself collapses onto the Galilean transformation of Section 1,

$$x' = \gamma\,(x - vt) \longrightarrow x - vt, \qquad t' = \gamma\Big(t - \frac{v}{c^2}\,x\Big) \longrightarrow t. \tag{30}$$

Relativity does not overturn Newtonian mechanics, then, so much as contain it: Newton's physics is the low-speed limit of Einstein's, recovered whenever $v \ll c$. This is why the strange effects of relativity went unnoticed for centuries — at the speeds of everyday life, γ is indistinguishable from 1.